Assignment 4

Visual Information Processing and Management

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1 Object Instance Detection

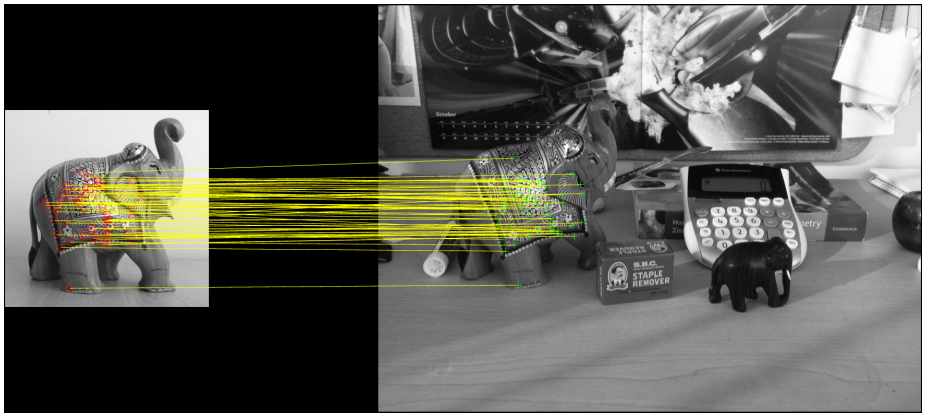


Figura 1: Match



Figura 2: Bounding Box



Figura 3: Contorno

2 Codice

```
% Lettura immagini
elephantImage = imread("immaginiObjectDetection/elephant.jpg");
sceneImage = imread("immaginiObjectDetection/clutteredDesk.jpg");

%% Keypoint detection
elephantPoints = detectSURFFeatures(elephantImage, "MetricThreshold
    ", 500, "NumScaleLevels", 6);
scenePoints = detectSURFFeatures(sceneImage, "MetricThreshold",
    500, "NumScaleLevels", 6);

figure(1), clf
imshow(elephantImage), hold on
plot(selectStrongest(elephantPoints, 100)), hold off

figure(2), clf
imshow(sceneImage), hold on
plot(selectStrongest(scenePoints, 100)), hold off

%% Keypoints description
[elephantFeatures, elephantPoints] = extractFeatures(elephantImage,
     elephantPoints, "FeatureSize", 128);
[sceneFeatures, scenePoints] = extractFeatures(sceneImage,
    scenePoints, "FeatureSize", 128);

%% Features matching
matches = matchFeatures(elephantFeatures, sceneFeatures, "
    MatchThreshold", 5, "MaxRatio", 0.9);
matchedElephantPoints = elephantPoints(matches(:, 1));
matchedScenePoints = scenePoints(matches(:, 2), :);
```

```
figure(3), clf
showMatchedFeatures(elephantImage, sceneImage,
    matchedElephantPoints, matchedScenePoints, "montage");

%% Geometric consistency check
[tform, inlierElephantPoints, inlierScenePoints] = ...
    estimateGeometricTransform(matchedElephantPoints,
    matchedScenePoints, "affine", "MaxDistance", 47);
figure(4), clf
showMatchedFeatures(elephantImage, sceneImage, inlierElephantPoints
    , inlierScenePoints, "montage")

size(inlierElephantPoints)

%% Bounding box drawing
boundigBox = [1 1;
           size(elephantImage, 2) 1;
           size(elephantImage, 2) size(elephantImage, 1);
           1 size(elephantImage, 1);
           1 1];
newBoundingBox = transformPointsForward(tform, boundigBox);
figure(5), clf
imshow(sceneImage), hold on
line(newBoundingBox(:, 1), newBoundingBox(:, 2), "Color", "y")
hold off

%% Precise bounding box
% figure(6), clf
% imshow(elephantImage)
% [x, y] = ginput(50);
% x = [x; x(1)];
% y = [y; y(1)];
% save elephantPoints x y

load elephantPoints.mat

newBoundingBox = transformPointsForward(tform, [x y]);
figure(7), clf
imshow(sceneImage), hold on
line(newBoundingBox(:, 1), newBoundingBox(:, 2), "Color", "y")
hold off
```